MILITARY SPECIFICATION

CENTRIFUGE, LABORATORY, 115 Volt, 50/60 Hz

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of table model centrifuge suitable for general laboratory use where the character of the work requires the performance of a wide variety of laboratory testing.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-585	- Boxes, Wood, Wirebound.
PPP-B-601	- Boxes, Wood Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116	- Preservation-packaging, Methods of
MIL-M-7298	- Manual, Technical, Commercial Equipment.
MIL-L-10547	- Liner, Case and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, Code ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6640

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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI C73.11	- Dimensions of Plugs and Receptacles - General Purpose, 125 Volts, 15 Amperes, 2-Pole, 3-Wire, Grounding Type.

(Application for copies should be addressed to the American Standards Institute, 1430 Broadway, New York, NY 10018.)

UNDERWRITERS LABORATORIES (UL)

UL - 73	- Standard for Safety, Electric Motor-Operated Appliances.

(Application for copies should be addressed to the Underwriters Laboratories Inc., 1285 Walt Whitman Road, Melville, L.I., NY 11746; 207 E. Ohio Street, Chicago, IL 60611; or 1655 Scott Blvd., Santa Clara, CA 95050.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material. Material shall be as specified herein. When material is not specified, it shall be of a quality which meets the functional requirements of the centrifuge. All materials shall be resistant to corrosion or suitably protected to prevent corrosion. Commercial tolerances shall apply for all thicknesses of cold rolled steel.

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3.1.1 Motor base. The base shall be fabricated from 0.1345 inch thick (No. 10 Manufacturers' Gage) cold rolled steel and shall be of such design and size that it regidly supports the housing assembly and the motor.

3.1.2 Housing. The housing shall be fabricated from 0.0359 inch thick (No. 20 Manufacturers' Gage) cold rolled steel.

3.1.3 Cover. The cover shall be fabricated from 0.0598 inch thick (No. 16 Manufacturers' Gage) cold rolled steel.

3.1.4 Head and trunnion rings. The head and trunnion rings shall be fabricated from cast brass.

3.1.5 Shields. The shields shall be fabricated from aluminum alloy.

3.1.6 Tube cushions. The tube cushions shall be fabricated from felt, rubber or synthetic rubber.

3.2 Design and construction. The centrifuge shall be as shown in fig. 1 for style and design and shall consist of a motor base, a horizontal head, a motor, motor controls and a housing with a cover. Accessories to be supplied with the centrifuge shall be as specified in 3.2.9. The overall size of the centrifuge shall be approximately 22 inches in diameter and 14 inches high. The centrifuge shall be capable of accommodating 1/2 to 100 milliliter (ml) tubes. The overall weight of the centrifuge including the head shall not exceed 70 pounds. The maximum speed, as a minimum, shall be 2250 revolutions per minute (rpm).

3.2.1 Base. The base shall be of sufficient weight to assure stability without creeping during operation. The drive motor and motor controls shall be mounted in the base. The motor mount shall be designed to eliminate vibration during operation.

3.2.2 Housing. The centrifuge shall be equipped with a housing to eliminate the hazard of injury from accidental contact with the whirling tubes. The housing shall be attached to the base from the inside. The housing shall be equipped with a circular cover, hinged across its diameter, one half opening for access to the interior, the other half permanently attached to the housing. A latch shall be provided to firmly hold the cover closed during operation.

3.2.3 Head. The head shall be a machined casting and shall be dynamically balanced. The head shall be of the four position type. The head shall be attached to the motor shaft by means of a knurled nut or similar fastener capable of holding the head securely during operation but will facilitate easy removal of the head. The head shall be capable of accommodating the trunnion rings and shields specified in table I.

3.2.4 Trunnion rings. The trunnion rings shall be supported by the head in such a manner to insure free swing during centrifuge operation. The trunnion rings shall accommodate the corresponding size shield specified in table I.

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3.2.5 Shields. Each shield shall have cushioning at the bottom to protect the tubes that are placed in the shields for centrifuging.

3.2.6 Motor. The motor shall be a variable speed motor designed for operation on 110-120 Volt, 50/60 Hz having a maximum power consumption of 330W. The motor assembly shall be suitable for continuous or intermittent duty in a vertical position, and shall have sufficient horsepower to drive a fully loaded head to a speed of 2250 revolutions per minute (rpm). The motor shall contain factory-lubricated grease-sealed ball bearings.

3.2.7 Motor controls.

3.2.7.1 Power switch. The centrifuge shall be provided with a heavy duty power switch for turning the centrifuge on and off. The switch shall be mounted in the base of the centrifuge with markings to indicate position of switch.

3.2.7.2 Speed control device. The centrifuge shall be provided with a speed control device which shall be of the continuous auto-transformer type. A calibrated reference dial shall be provided which gives accurate settings for the desired speed. The indicated dial setting shall be within $\pm$ 5 percent of the actual speed of the head when tested in accordance with 4.5.4. The speed control device shall be placed adjacent to the power switch.

3.2.8 Power cord. The power cord shall be a heavy duty, flexible, 3-conductor cable that terminates with a non-detachable, 3-blade male plug suitable for use with receptacles specified in ANSI C73.11. A strain relief bushing shall be provided where the power cord enters the base. The cord shall have a minimum external length of five feet. The terminal connections inside the housing shall show no evidence of strain when tested as specified in 4.5.1. The ground lead shall be so connected that all exposed metal parts of the centrifuge are electrically grounded when tested as specified in 4.5.2.

3.2.9 Accessories. The centrifuge shall include the following accessories specified in table I.

TABLE I. Accessories supplied with centrifuge.

Quantity	Accessory
28	Shield for 15 ml, round bottom tube
8	Shield for 50 ml, tapered bottom tube
4	Shield for 100 ml, cone-shaped tube
4	Shield for 100 ml, pear-shaped tube
4	Trunnion ring for seven 15 ml tubes
4	Trunnion ring for two 50 ml tubes
4	Trunnion ring for one 100 ml cone-shaped tube
4	Trunnion ring for one 100 ml pear-shaped tube

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3.3 Performance.

3.3.1 Relative centrifugal force (RCF). The design and dimensions of the head and trunnions shall be such that for the given speed and tube size the relative centrifugal force (RCF) indicated in table II shall be attained when calculated as specified in 4.5.6.

TABLE II. Relative centrifugal force.

Tube size	Speed (Revolutions per minute)	RCF
15 ml	2000	885
50 ml	2000	960
100 ml (pear-shaped)	2000	1210
100 ml (cone-shaped)	1700	780

3.3.2 Dielectric strength. There shall be no electrical breakdown when the centrifuge is subjected to the test specified in 4.5.3.

3.3.3 Continuous operation. The centrifuge shall be capable of continuous operation at any speed up to and including the maximum speed (3.2). When tested as specified in 4.5.5, the tubes shall remain in place, there shall be no damage to the centrifuge, and there shall be no excessive vibration or creeping. The head speed shall not vary more than plus or minus 5 percent of dial setting.

3.4 Finish. The interior and exterior of the housing, cover and base of the centrifuge and the exterior of the motor housing shall be primed, painted and baked. The primer and paint shall be applied and baked in such a manner as to provide a uniform, chip-proof and corrosion resisting surfaces. The manufacturer's commercial color for the finish shall be acceptable.

3.5 Product marking.

3.5.1 Identification marking. Each centrifuge shall have a nameplate of corrosion-resisting metal, securely mounted in a conspicuous location, and permanently and legibly marked with the letters U.S. and with the following information:

a. Manufacturer's name or registered trademark.
b. Model number.

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c. Serial number
d. Voltage
e. Frequency
f. Power requirements
g. Maximum operating speed

3.5.2 Head marking. The positions on the head shall be permanently numbered one through four.

3.6 Service data. The contractor shall furnish with each unit three copies each of a technical manual which contains complete instructions for installation, operation and maintenance, and lists of component parts. The manual shall comply with the requirements of MIL-M-7298. As soon as practicable after award of the contract, the contractor shall furnish to the contracting officer, via the cognizant Government representative, two preliminary copies of the proposed technical manual for review and comment by the Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary manuals shall be submitted sufficiently early to permit adequate review by the Directorate of Medical Materiel, Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of any required revisions and corrections in the final manual without delaying delivery under the contract. In addition, the contractor shall furnish 15 copies of the approved technical manual to the contracting officer for distribution. Should the supplier have furnished acceptable manuals for the identical item within two years preceding the date of contract and propose to furnish manuals identical with those previously accepted, the requirements for review of drafts and submission of 15 additional copies may be waived upon request to the contracting officer. Waiver of submission of 15 additional copies shall be at the option of the Government.

3.6.1 The data supplied shall include graphs, nomagraph or similar data for rapid conversions of revolutions per minute and known radii into relative centrifugal force (RCF).

3.7 Workmanship. Workmanship shall be first class throughout. The centrifuge and components shall be free from defects which detract from their appearance, or may impair their serviceability, or may endanger the safety of the operator. All metal components shall be free of burrs, sharp edges and deformities.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements, as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the materials may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table III. The unit of product for examination purposes shall be one centrifuge with accessories.

TABLE III. Sampling for examination.

Categories	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.0
Minor defects	II	2.5
For dimensional examination	S-1	2.5

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4.2.2 For tests. Sampling for tests shall be conducted in accordance with table IV. The unit of product for test purposes shall be one centrifuge. Acceptance number shall be zero for all sample sizes.

TABLE IV. Sampling for tests.

Characteristic	Requirement	Test procedure	Inspection level
Power cord strain	3.2.8	4.5.1	100%
Continuity of ground	3.2.8	4.5.2	100%
Speed control dial	3.2.7.2	4.5.4	100%
Dielectric strength	3.3.2	4.5.3	100%
Continuous operation	3.3.3	4.5.5	100%
Relative centrifugal force	3.3.1	4.5.6	100%

4.3 Examination.

4.3.1 Centrifuge. The centrifuge shall be examined to determine compliance with all the requirements contained in this specification. Examination shall be conducted as specified in tables III and V. Examination however, shall not be restricted to defects listed in table V.

TABLE V. Classification of defects.

Classification	Defect
Major	
101	Parts or accessories missing.
102	Parts or accessories not usable or not capable of being properly secured to unit.
103	Metal parts not free of corrosion.
104	Metal parts not free of burrs or sharp edges.
105	Centrifuge not free of wobbling or rocking when placed on a flat surface.
106	Cover not adequately attached to housing.
107	Cover not capable of being easily opened, closed or securely latched.
108	Head not capable of being easily secured to or removed from motor shaft.
109	Internal wiring not adequately secured to terminations.

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TABLE V. Classification of defects.(continued).

Classification	Defect
Minor	
201	Finish not free of scratches, chipping, cracking, or peeling.
202	Identification marking not complete, correct, legible or permanent.
203	Unit not free of dirt, grease, or foreign matter.

4.3.2 Dimensional examination. The component parts shall be examined for defects in dimensions. Any dimension not within the tolerances specified herein shall be classified as a defect.

4.4 Packaging inspection. The inspection of the unit package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.5 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.5.1 Power cord strain. The power cord strain test shall be conducted in accordance with UL-73.

4.5.2 Continuity of ground. A continuity indicating device shall be used to check continuity between the ground pin on the input plug at the end of the power cord and the base, head, housing and cover of the centrifuge. Grounding procedures shall be in accordance with UL-73.

4.5.3 Dielectric test. The dielectric test shall be performed in accordance with UL-73.

4.5.4 Speed control dial. Install four 100 milliliter water filled tubes in the head of the centrifuge and operate the centrifuge with the cover open. Set the speed control dial at each successive 200 RPM setting, in turn, and check the head speed by means of a strobotac or tachometer to determine compliance with 3.2.7.2.

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4.5.5 Continuous operation. With the head loaded as specified in 4.5.4 and the cover closed, operate the centrifuge for 24 hours at maximum rheostat setting. The centrifuge shall comply with the requirements of 3.3.3.

4.5.6 Relative centrifugal force (RCF). The head shall be placed on the drive shaft with the appropriate trunnion ring for the tubes specified in table II. The radius of rotation shall be measured from the center of the drive shaft to the center of gravity of the tube in the trunnion ring. Using the radius measured and the speed specified in table II calculate the RCF as indicated below. The RCF shall be calculated for each tube size and speed indicated.

$$RCF = 28.6 \times N^2 \times R$$

N = revolutions per minute in thousands
R = radius in inches to center of gravity of tube

5. Packaging.

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit packaging. Each centrifuge, with parts, and accessories, shall be packaged Method 11b of MIL-P-116. The interior box shall be of appropriate size and design constructed in accordance with PPP-B-636, Type CF, class domestic. The box shall be designed with suitable interior fittings or shall contain sufficient cushioning materials to ensure against shifting of, or physical damage to the contents. Closure of the box shall be in accordance with the appendix to the box specification. The exterior container shall conform to the level of pack specified (see 5.2).

5.1.1.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. One (1) centrifuge shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard box shall be waterproofed in accordance with the appendix of PPP-B-636.

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5.2.2 Level B. One (1) centrifuge shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Unit and exterior containers. Each unit package and exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The centrifuge covered by this specification is intended to be used in laboratories to perform a variety of tests.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

a. Title, number and date of this specification.
b. National stock number (NSN).
c. Applicable levels of packaging and packing. (see 5.1 and 5.2)

6.3 This specification covers the following item(s) appearing in the Federal Supply Catalog:

National Stock Number	Item Identification
6640-00-062-4336	CENTRIFUGE, LABORATORY, 115 volt, 50-60 Hz.

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6.4 Accessory spare parts. Accessories for the centrifuge covered by this specification (3.2.9) are supported by the following spare parts which are carried as separate line items in the Department of Defense supply system. It should be noted that the centrifuge (6640-00-062-4336) includes all of the accessory parts listed below.

6640-00-762-2814 - Shield for 50 ml tapered bottom tubes.
6640-00-360-1062 - Shield for 100 ml cone-shaped tubes.
6640-00-359-9951 - Shield for 100 ml pear-shaped tubes.
6640-00-629-1652 - Trunnion ring for two 50 ml tubes.
6640-00-690-3269 - Trunnion ring for one 100 ml cone-shaped tube.
6640-00-690-3282 - Trunnion ring for one 100 ml pear-shaped tube.

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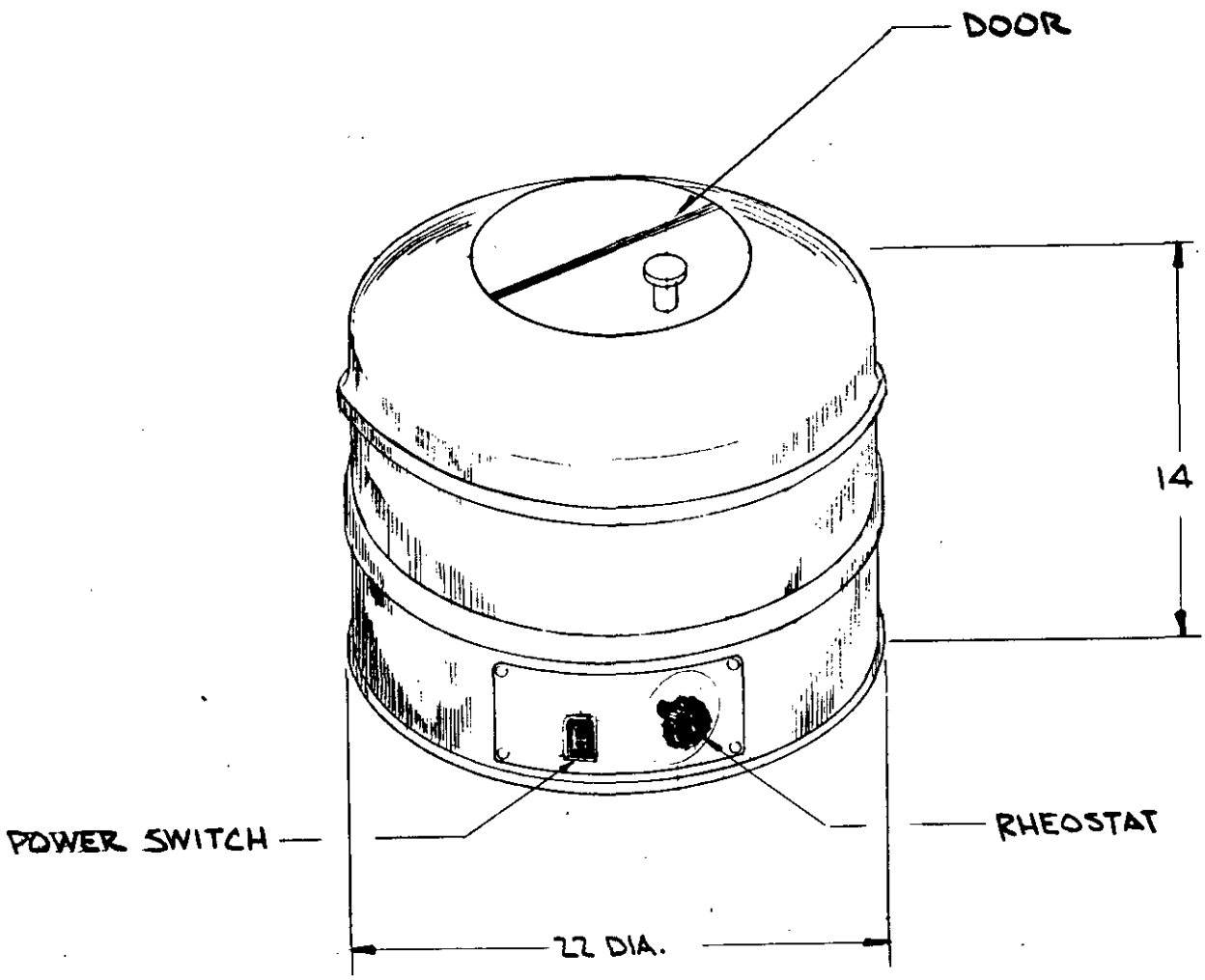


FIGURE 1. Centrifuge, laboratory.

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MIL-C-37220 - CENTRIFUGE, LABORATORY

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A. GIVE PARAGRAPH NUMBER AND WORDING.

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